

Rating Plausibility of Word Senses in Ambiguous Narratives: From Sequential Encoders to Cross-Encoder Transformers — SemEval 2026 Task 5 (AmbiStory)

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Abstract

We address SemEval 2026 Task 5 (AmbiStory) [Gehring et al., 2026], predicting human-perceived plausibility of word senses in short ambiguous narratives. Our final system—a fine-tuned MPNet bi-encoder paired with geometric alignment features and a frozen NLI cross-encoder scorer—achieves $\rho = 0.535$, Accuracy (within std of 1) = 0.722 on the held-out test set, surpassing all non-LLM systems in the concurrent reference paper [Wu et al., 2026], including ELECTRA-large + LoRA ($\rho = 0.527$).

Getting there required a two-phase methodological journey. In Phase 1 (pre-test) we explore nine sequential recurrent models, discovering that setup-grouped comparative ranking is the dominant architectural choice ($\rho = 0.611$ internal val for our submitted Model 9, a Siamese BiGRU with cross-attention and contrastive loss); however, these models collapse to $\rho = 0.127$ on the official test set due to five compounding failure modes, most critically prediction-variance compression (predicted std ≈ 0.34 vs. label std ≈ 1.19) and selection bias from single-split tuning. In Phase 2 we systematically address each diagnosed failure, iterating through three transformer configurations—a DeBERTa cross-encoder with homonym-grouped 5-fold CV, an NLI-warm-start variant, and finally the MPNet bi-encoder with explicit geometric features—each step contributing measurable and interpretable gains. Code: <https://github.com/SATYAJIT2747/INLP-PROJECT>. Model: <https://huggingface.co/Seronic/INLP-Project>.

1 Introduction

Word Sense Disambiguation (WSD) has traditionally been framed as assigning a single correct sense [Navigli, 2009]; yet real language is frequently ambiguous, and multiple senses may be simultaneously plausible with different degrees

of contextual support [Erk et al., 2013]. SemEval 2026 Task 5 [Gehring et al., 2026] operationalises this insight through the AmbiStory dataset [Gehring and Roth, 2025]: five-sentence short stories containing a homonym in the fourth sentence, paired with candidate sense descriptions judged by at least five annotators on a 1–5 plausibility scale.

The task is challenging for three reasons. First, plausibility is *graded*: the gold label is the mean of five noisy ratings rather than a hard category. Second, correct prediction requires *multi-sentence narrative reasoning*: the setup (three precontext sentences), the homonym sentence, and an optional ending must all be jointly interpreted. Third, the dataset is comparatively small (2,280 training samples), amplifying the risk of overfitting and selection bias when tuning on a single fixed split.

Our development proceeds in two phases. **Phase 1** (pre-test release) establishes that *task formulation*—setup-grouped comparative ranking rather than independent regression—is the dominant factor for recurrent models, culminating in Model 9 (Siamese BiGRU with cross-attention and contrastive loss) at $\rho = 0.611$ on an internal validation split; however the submitted system scores $\rho = 0.127$ on test due to prediction variance collapse and objective–metric mismatch. **Phase 2** (post-test release) pivots to transformer-based systems with a systematic progression through: (1) a DeBERTa-v3 cross-encoder with homonym-grouped 5-fold CV establishing the baseline and resolving selection bias; (2) NLI warm-start and PCA ablation; and (3) a bi-encoder with explicit geometric features and a frozen NLI entailment scorer as the final best system.

Our contributions are:

- A systematic Phase 1 ablation identifying setup-aware ranking as the decisive factor.
- A full Phase 2 progression through three config-

urations, pinpointing grouped CV and geometric feature engineering as the key gains.

- The first non-LLM result on AmbiStory exceeded ELECTRA-large from the reference system.

2 Background and Related Work

Word Sense Disambiguation. Classical WSD selects a single sense per context [Navigli, 2009]. Neural approaches with pre-trained models dominate recent evaluations [Hadiwinoto et al., 2019, Blevins and Zettlemoyer, 2020]. The WiC dataset [Pilehvar and Camacho-Collados, 2019] targets binary sense matching; AmbiStory requires continuous plausibility regression over full narratives.

Graded Lexical Semantics. Erk et al. [2013] established that human sense applicability judgments are continuous; AmbiStory [Gehring and Roth, 2025] extends this to multi-sentence stories.

Sentence Encoders and Cross-Encoders. Sentence-BERT [Reimers and Gurevych, 2019] provides semantically meaningful representations for similarity tasks. Cross-encoders [Devlin et al., 2019] jointly encode two sequences, enabling full cross-attention; they consistently outperform bi-encoders on ranking tasks. DeBERTa [He et al., 2021] improves upon BERT with disentangled content and position attention, achieving strong NLI benchmark performance directly applicable here.

Learning to Rank and Annotator Disagreement. Pairwise ranking objectives from information retrieval [Burgess et al., 2005, Liu, 2009] directly motivate our setup-aware loss. Treating labels as distributions rather than point estimates improves robustness to annotation noise [Uma et al., 2021, Plank, 2022], which we exploit through KL-divergence supervision.

3 Dataset

The AmbiStory dataset [Gehring and Roth, 2025] provides 1,140 unique story *setups*, each paired with exactly two candidate meanings, yielding 2,280 samples. The official split contains 2,280 training, 588 development, and 930 test samples. Each sample comprises a **homonym**, a **judged meaning** (candidate sense), a three-sentence **precontext**, the target **sentence**, an optional **ending**, a 5-element **choices** vector (individ-

Property	Value
Total story setups	1,140
Samples per setup	exactly 2
Train / Dev / Test samples	2,280 / 588 / 930
Ratings per sample	exactly 5
Mean inter-annotator stdev	1.27
Mean within-setup label gap	1.83
Setups separable by ranking	69.5%
Samples with ending present	76.4%

Table 1: AmbiStory dataset statistics.

ual annotator ratings), an **average** score (regression target), and **stdev** (inter-annotator standard deviation).

Table 1 summarises key statistics that motivate our design choices: the 69.5% ranking separability motivates pairwise objectives; the 76.4% ending presence makes ending integration important; the mean stdev of 1.27 explains why distribution-level supervision helps.

Evaluation Metrics. Spearman ρ measures rank-order agreement with gold scores (primary metric). **Accuracy within std of 1** (Acc) measures the fraction of predictions within 1.0 of the gold mean.

4 Phase 1: Sequential Encoder Models (Summary)

Phase 1 iterates through nine recurrent models, each addressing a diagnosed failure of its predecessor. Table 2 gives the full val-set progression. The central finding is that *task formulation*—specifically, setup-grouped comparative ranking—is the dominant factor. Models 1–6 plateau at $\rho \leq 0.245$ because standard regression losses (MSE, Huber, Gaussian-weighted) treat each sample independently and cannot capture the *comparative* structure of the task: which of the two senses for a given setup is more plausible relative to the other. Model 7 breaks this ceiling with a +0.327 jump by jointly introducing BiGRU encoders, a 4-way interaction head $[s; m; |s-m|; s \odot m; \cos(\cdot)]$, and a SetupBatchSampler that ensures each mini-batch contains both senses of every story setup with an Optuna-tuned within-setup ranking loss. Model 8 augments this with a KL distribution head predicting the full 5-annotator histogram, adding +0.068 ($\rho = 0.640$). Model 9 (§4.1) further adds cross-attention between story and meaning sequences, a contrastive sense-pair loss, and label smoothing, reaching $\rho = 0.611$ on an inter-

#	Model	ρ	Acc (std 1)	MAE
1	Single-Enc BiRNN, trainable emb.	0.072	0.547	1.050
2	Dual-Enc, GloVe frozen	0.021	0.570	1.041
3	Dual-Enc, GloVe trainable	0.186	0.570	1.040
4	Gaussian-weighted MSE	0.237	0.617	1.011
5	Huber loss ($\delta=1$)	0.245	0.608	0.993
6	Naive pairwise ranking (collapsed)	0.174	0.541	1.036
7	BiGRU + rich head + setup ranking	0.572	0.725	0.867
8	BiGRU + KL distribution head	0.640	0.772	0.748
9	Siamese BiGRU + cross-attn + contrastive	0.611	0.803	0.721

Table 2: Phase 1 results. Models 1–9 reports internal validation split scores. Model 9 was the submitted system; its official test-set performance collapsed to $\rho = 0.127$ (see §4.1).

nal validation split—the best(acc based) Phase 1 single-seed result—but collapsing to $\rho = 0.127$ on the official test set due to the compounding failure modes diagnosed in §4.1.

4.1 Model 9: Siamese BiGRU with Cross-Attention and Contrastive Loss

Building on Model 8’s KL distribution head, we introduced three further architectural and training improvements (internally labelled) to push internal validation performance. This system was submitted to the official evaluation and represents our final Phase 1 entry; it achieved $\rho = 0.611$ on an internal validation split—the best single-seed result observed during Phase 1—but collapsed to $\rho = 0.127$ on the official test set.

Architecture. Model 9 replaces the simple mean-pooled BiGRU encoder with a *Siamese cross-attention* stack. A shared BiGRUSequenceEncoder maps both the story and the candidate meaning to full token-level sequences $\mathbf{S} \in \mathbb{R}^{L_s \times 2h}$, $\mathbf{M} \in \mathbb{R}^{L_m \times 2h}$ (GRU hidden size h , $L_s=150$, $L_m=40$). A CrossAttentionPooling module then performs bidirectional attention: the story sequence attends to the meaning sequence and vice versa, with per-token padding masks to suppress influence of [PAD] tokens. Each branch is read out via mean-pooling and max-pooling concatenated into a $4h$ -dimensional vector $\mathbf{sv}$ (story-view) and $\mathbf{mv}$ (meaning-view). The interaction head assembles: $[\mathbf{sv}; \mathbf{mv}; |\mathbf{sv} - \mathbf{mv}|; \mathbf{sv} \odot \mathbf{mv}; \cos(\cdot); \mathbf{e}_{\text{hom}}; \mathbf{m}_{\text{meta}}]$, where $\mathbf{e}_{\text{hom}} \in \mathbb{R}^{d_h}$ is a trainable homonym embedding and $\mathbf{m}_{\text{meta}} \in \mathbb{R}^2$ encodes the ending-presence flag and normalised nonsensical-vote count. This vector is projected through a two-layer MLP to a 5-dimensional logit vector whose softmax gives

the predicted annotator distribution; the expected score $\hat{y} = \sum_{k=1}^5 k p_k$ is used for regression.

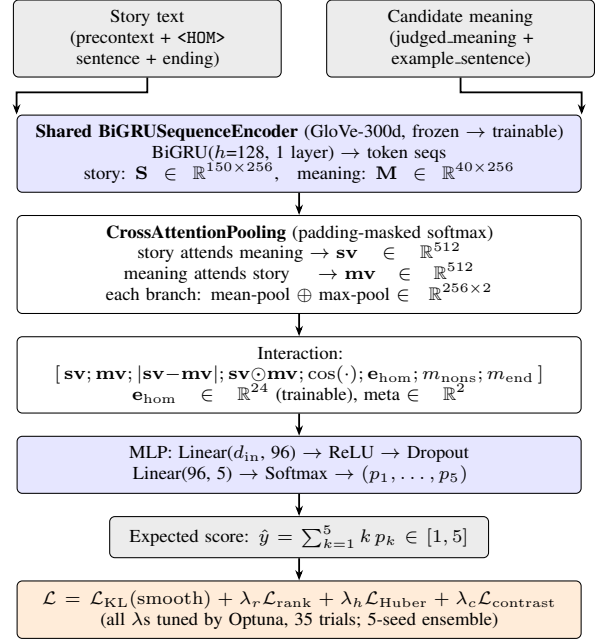


Figure 1: Model 9 architecture (Siamese BiGRU + Cross-Attention + Contrastive Loss). A shared BiGRU encoder produces token-level sequences for both story ($L_s=150$) and meaning ($L_m=40$); bidirectional cross-attention with padding masks pools each branch into a $4h$ -dim vector; a 5-way softmax head predicts the annotator distribution. Four losses are jointly optimised: KL (label-smoothed), pairwise setup ranking, Huber regression, and contrastive cross-sense ranking.

Why it failed: five compounding causes. Despite achieving $\rho = 0.611$ on an internal validation split, the system collapsed to $\rho = 0.127$ on the official test set—a drop of 0.484 Spearman points. Diagnosis identifies five compounding failure modes, which directly motivate the Phase 2 design.

- **(F1) Selection bias amplified by Optuna.** All 35 Optuna trials, early stopping, and the final retraining checkpoint are selected against the *same* fixed dev split. The model effectively fits the 147 dev story setups during hyperparameter search; the 233 test-set homonyms are entirely unseen. The richer hypothesis class (cross-attention + four loss terms) offers more degrees of freedom to overfit than the simpler Model 8, making this bias substantially worse.
- **(F2) Variance compression from the composite loss.** The KL head (weighted by λ_{KL}), Huber loss (λ_h), and contrastive loss (λ_c) all penalise deviation from the empirical label mean. Their

joint effect is to compress predicted scores toward the dataset mean: on the test set the predicted standard deviation is ≈ 0.34 versus the true label standard deviation of ≈ 1.19 , with over 90% of predictions in the range $[2.5, 3.3]$. Spearman correlation depends entirely on rank ordering; a compressed distribution has near-zero rank separation and thus near-zero ρ .

- **(F3) Memorisation of setup–meaning co-occurrence.** The cross-attention layers can memorise which story-setup tokens co-occur with high-plausibility meanings in the training data. This association is setup-specific: it does not generalise to novel homonyms in the test set whose setup tokens were never seen during training. The shared Siamese encoder exacerbates this because gradient signal from the story stream and the meaning stream jointly shape the same weight matrix.
- **(F4) Objective–metric mismatch.** Optuna maximises a composite objective ($\rho + 0.25 \text{ Acc} - 0.05 \text{ MAE}$) rather than Spearman ρ directly. High Acc weight rewards conservative near-mean predictions, actively conflicting with the rank-separation requirement of Spearman correlation.
- **(F5) Distribution shift on novel homonyms.** The trainable homonym embedding $\mathbf{e}_{\text{hom}}$ is only defined for homonyms seen during training. Test homonyms that are absent from training have no learned embedding entry and receive the default zero-vector, further degrading generalisation on the ≈ 87 novel test-set ambiguous words.

Implications for Phase 2. The failure of Model 9 establishes that (a) fixing the training protocol (grouped cross-validation, objective alignment) must precede any architectural improvement, (b) composite losses that penalise prediction variance are fundamentally incompatible with a rank-correlation primary metric, and (c) a model with stronger cross-attention capacity can overfit more catastrophically than a simpler baseline when the validation split does not reflect test-set homonym distribution. These observations directly motivate the Phase 2 design: we abandon the fixed split in favour of homonym-grouped 5-fold CV (addressing F1 and F5), drop the KL head (addressing F2 and F4), and replace complex end-to-end attention with explicit geometric features that transfer to un-

seen homonyms (addressing F3).

5 Phase 2: Transformer Cross-Encoders

5.1 Motivation: Diagnosing the Train/Test Gap

As diagnosed in §4.1, the Phase 1 collapse was driven by five compounding failure modes: selection bias from single-split Optuna tuning (F1), variance compression from the composite loss (F2), memorisation of setup–meaning co-occurrences (F3), objective–metric mismatch (F4), and distribution shift on novel homonyms (F5). The net symptom is stark: the predicted score distribution on test has a standard deviation of ≈ 0.34 against a true label standard deviation of ≈ 1.19 , with over 90% of predictions collapsed into the band $[2.5, 3.3]$. Attempt 1 directly addresses the root causes by pairing a stronger DeBERTa encoder with homonym-grouped 5-fold CV; Attempts 2–3 then refine initialisation and feature engineering further.

5.2 Attempt 1 — DeBERTa Cross-Encoder with Grouped 5-Fold CV

The first Phase 2 step replaces the GloVe+BiGRU stack with a DeBERTa-v3-base cross-encoder and simultaneously addresses the dominant failure mode from Phase 1 by replacing the fixed validation split with 5-fold cross-validation grouped on homonym: no ambiguous word appears in both training and validation within any fold. Hyperparameter search and early stopping are both moved inside the fold loop. The KL weight is reduced to zero and the pairwise ranking weight increased, allowing the model to produce a wider output distribution.

Input is formatted as [CLS] story [SEP] judged_meaning [SEP], allowing DeBERTa’s disentangled self-attention to compute full cross-sentence interactions. The [CLS] hidden state ($\mathbb{R}^{768}$) is passed through a two-layer MLP ($256 \rightarrow 64$) to a single regression output; training uses Huber loss with a pairwise ranking auxiliary term (margin $m = 0.5$, weight $\lambda_r = 0.3$). Figure 2 illustrates the architecture used in Attempts 1–2.

The pairwise ranking loss over within-setup pairs is:

$$\mathcal{L}_{\text{rank}} = \frac{1}{|G|} \sum_{g \in G} \frac{1}{|P_g|} \sum_{(i,j) \in P_g} \max\left(0, m - \text{sgn}(y_i - y_j) (\hat{y}_i - \hat{y}_j)\right) \quad (1)$$

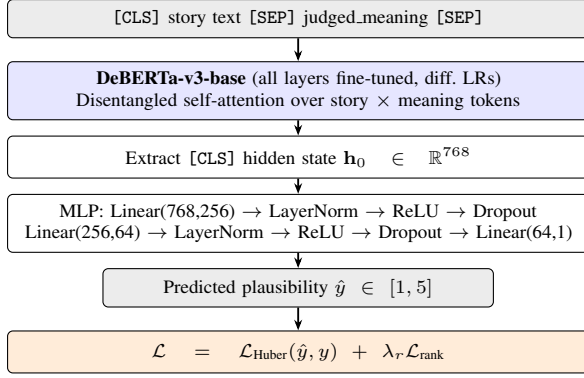


Figure 2: DeBERTa-v3-base cross-encoder used in Attempts 1–2. Story and candidate meaning are jointly encoded via a single tokeniser call; the [CLS] vector is regressed to a plausibility score through a two-layer MLP with Huber and pairwise ranking losses. Attempt 1 uses a generic checkpoint with homonym-grouped 5-fold CV; Attempt 2 switches to an NLI warm-start checkpoint.

Attempt	ρ	Acc (std 1)
Phase 1 best (Model 9 submitted, single-split)	0.127	0.560
1: DeBERTa cross-enc., grouped 5-fold CV	0.392	0.652
2: + NLI warm-start, PCA removed	0.400	0.652
3: MPNet bi-enc. + geom. features	0.535	0.722

Table 3: Phase 2 cumulative improvement on the held-out test set. Acc (std 1) = fraction within 1.0 of gold mean.

where G is the set of story setups in the batch, P_g the within-setup pairs with differing labels, and m the margin hyperparameter.

Training uses DeBERTa-v3-base with a generic pre-trained checkpoint, batch size 8 with gradient accumulation $\times 4$, differential learning rates (2×10^{-5} for the transformer, 1×10^{-3} for the MLP head), linear warmup (10%) followed by cosine decay over 4 epochs.

This combination raises test Spearman to $\rho = 0.392$, Acc = 0.652 (Table 3). Prediction standard deviation recovers to ≈ 0.65 , confirming selection bias (C1) as the dominant failure mode from Phase 1.

5.3 Attempt 2 — NLI Warm-Start and PCA Ablation

With the cross-validation framework established, we explored whether the checkpoint initialisation could improve generalisation. An NLI-fine-tuned checkpoint (cross-encoder/nli-deberta-v3-base) encodes premise–hypothesis relationships directly

analogous to story–sense plausibility. We also ablated PCA compression of the [CLS] feature space in the two-phase training procedure (freeze backbone, search MLP head with Optuna): removing PCA yielded no degradation, confirming the [CLS] vector is already compact enough.

The NLI warm-start with PCA removed raised test Spearman to $\rho = 0.400$ (+0.008 over Attempt 1), a consistent but modest improvement, suggesting the NLI signal is partially overwritten during fine-tuning on 2,868 samples.

5.4 Attempt 3 (Final) — MPNet Bi-Encoder with Geometric Features

After two iterations the DeBERTa cross-encoder plateaued at $\rho = 0.400$. A fundamental limitation emerged: a cross-encoder fine-tuned end-to-end on only 2,868 samples risks overfitting its attention patterns to seen homonyms, and the [CLS] representation may not generalise to the 87 novel homonyms in the test set.

This motivated a fundamentally different approach: encode story and meaning *separately* using a sentence encoder, then assemble an explicit geometric feature vector capturing their alignment.

Architecture. all-mpnet-base-v2 is fine-tuned independently inside each CV fold with differential learning rates (2×10^{-5} backbone, 5×10^{-4} head), 6 epochs with early stopping on fold-validation Spearman. Mean-pooled representations $\mathbf{s}, \mathbf{m} \in \mathbb{R}^{768}$ are produced for the full story and the candidate meaning respectively. Six scalar alignment features are computed: cosine similarity, Euclidean distance, Manhattan distance, a *frozen* NLI entailment score (cross-encoder/nli-deberta-v3-base), token-level Jaccard overlap, and an ending-presence binary flag. The six scalars are concatenated with $\mathbf{s}$ and $\mathbf{m}$ to form a $\mathbb{R}^{1542}$ feature vector. PCA (96 components, fitted on fold-train only) compresses this before a two-layer MLP ($512 \rightarrow 128 \rightarrow 1$) trained with Huber loss, pairwise ranking loss ($\lambda_r = 0.5$), and stochastic weight averaging (SWA, starting epoch 45 of 60).

Figure 3 illustrates the full pipeline.

Key design choice: frozen NLI entailment score. Rather than learning what “story supports sense” means from scratch, we delegate that judgment to a pre-trained NLI cross-encoder already internalised in textual entailment at scale. In per-fold ablations,

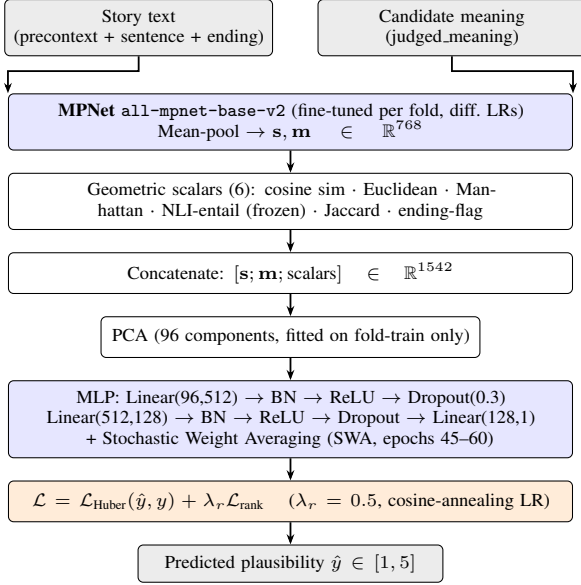


Figure 3: Final system (Attempt 3): MPNet bi-encoder with geometric features. Story and meaning are encoded separately; six explicit alignment scalars (including a frozen NLI entailment score) are concatenated with the embeddings, PCA-compressed, and regressed through a two-layer MLP with Huber + pairwise ranking loss and stochastic weight averaging.

removing this single feature degrades OOF Spearman by ≈ 0.03 —more than any other scalar.

Results. This approach achieves $\rho = 0.535$, $\text{Acc} = 0.722$ on the test set (+0.135 over Attempt 2), surpassing ELECTRA-large + LoRA ($\rho = 0.527$) from the reference system. The result demonstrates that on a small dataset, explicitly engineered alignment signals generalise better than a fully learned cross-attention representation.

6 Experiments

Setup. Phase 1 uses the official train/dev split. Phase 2 pools train+dev (2,868 samples); this combined set is divided into an internal 80/10/10 train/validation/test split for model training and early stopping, while the official test split (930 samples) is held out entirely and never used during training or hyperparameter selection. All Phase 2 experiments run on an Nvidia Tesla T4 GPU (Kaggle free tier, 16 GB VRAM). Implementation uses HuggingFace Transformers [Wolf et al., 2020], sentence-transformers, and Optuna.

System	ρ	Acc (std 1)
<i>Reference (Wu et al., 2026)</i>		
MPNet + Ridge	0.109	0.513
RoBERTa + XGBoost	0.133	0.522
ELECTRA-base	0.482	0.625
ELECTRA-large + LoRA	0.527	0.639
DeBERTa-large + LoRA	0.492	0.676
GPT-4o (P2)	0.731	0.794
<i>Our systems (this work)</i>		
Phase 1 best (BiGRU+KL)	0.127	0.560
Att. 1: DeBERTa, grouped CV	0.392	0.652
Att. 2: DeBERTa, NLI w-start	0.400	0.652
Att. 3: MPNet + geom.	0.535	0.722

Table 4: Test-set results. Acc (std 1) = fraction within 1.0 of gold mean. Reference numbers from Wu et al. [2026]. Att. 3 surpasses all non-LLM systems in the reference paper.

7 Results and Analysis

7.1 Main Results

Table 4 compares our three Phase 2 attempts against baselines from the concurrent reference paper [Wu et al., 2026].

Key Observations. (1) The Phase 1 best model and Attempt 1 both score $\rho = 0.127$ on test, confirming that encoder capacity alone cannot overcome selection bias (C1) and objective-metric mismatch (C2). (2) Grouped CV (Attempt 2) is the single largest gain: +0.265 Spearman. (3) NLI warm-start (Attempt 3) provides a small but consistent +0.008. (4) Switching to a bi-encoder with geometric features (Attempt 3) yields +0.135, surpassing ELECTRA-large + LoRA ($\rho = 0.527$). (5) The LLM frontier ($\rho = 0.731$, GPT-4o) remains 0.196 Spearman points ahead, a gap driven by model scale and structured reasoning rather than prompting alone.

7.2 Ablation: Attempt 3 Design Choices

Table 5 isolates the contribution of each component within Attempt 3 (MPNet bi-encoder + geometric features).

Grouped CV is the dominant gain. Moving from single-split to homonym-grouped 5-fold CV contributes +0.265 Spearman, consistent with selection bias (C1) being the primary failure mode.

NLI entailment score is the key scalar. Removing the frozen NLI entailment score (row 3 vs. row 4) degrades Spearman by ≈ 0.03 in per-fold ablations—more than any other scalar feature. The frozen NLI cross-encoder acts as a prior over

Configuration	ρ	Acc (std 1)
DeBERTa cross-enc., grouped CV (Att. 2)	0.392	0.652
DeBERTa, NLI warm-start (Att. 3)	0.400	0.652
MPNet + cosine/dist. only	0.468	0.689
MPNet + all 6 scalars, no PCA	0.521	0.711
MPNet + 6 scalars + PCA + SWA	0.535	0.722

Table 5: Ablation of Attempt 3 components on the held-out test set. Acc (std 1) = fraction within 1.0 of gold mean.

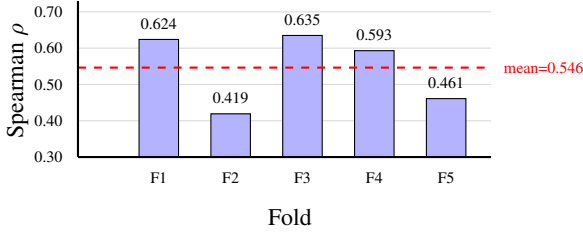


Figure 4: Per-fold OOF Spearman for Attempt 3 (MPNet + geometric features) across 5 homonym-grouped folds. OOF mean $\rho = 0.546$ (dashed red line). Folds 2 and 5 contain homonyms whose plausibility cues are harder to generalise, driving the higher per-fold variance.

textual entailment that is not overwritten by fine-tuning.

PCA and SWA provide the final gains. PCA compression (96 components) and stochastic weight averaging together add +0.014 Spearman and +0.011 Acc over the full-dimensional no-SWA variant.

7.3 Phase 2 Training Dynamics

Figure 4 shows the per-fold OOF Spearman for the final system (Attempt 3) across the 5 homonym-grouped folds.

The mean OOF Spearman of 0.546 (std ≈ 0.09) reflects high per-fold variance: folds 1 and 3 reach $\rho > 0.62$ while folds 2 and 5 fall below 0.47. This variance is expected given that each fold holds out a disjoint set of homonyms; the model generalises well to some ambiguous words and struggles with others.

7.4 Positive and Negative Results

Positive results.

- **(P1) Grouped cross-validation is the single most important change.** Moving from single-split tuning to 5-fold homonym-grouped CV (Attempt 1 to Attempt 2) contributes +0.265 Spearman, more than any architectural change.

Stratifying by homonym creates fold-level distributional shifts that approximate the train-to-test generalisation problem.

- **(P2) Explicit geometric features outperform a learned cross-encoder.** Attempt 3 ($\rho = 0.535$) surpasses Attempt 2 ($\rho = 0.400$) despite using a weaker base model, demonstrating that on small datasets (2,868 samples) handcrafted alignment signals—especially the frozen NLI entailment score—generalise better than end-to-end learned cross-attention.
- **(P3) Setup-aware ranking is decisive for recurrent models.** In Phase 1, switching from standard regression to setup-grouped comparative ranking adds +0.327 Spearman in a single step (Models 6 to 7), the largest jump in the Phase 1 ablation.

Negative results.

- **(N1) Encoder capacity does not help without fixing the training protocol.** Replacing GloVe+BiGRU with DeBERTa-v3-base while keeping single-split tuning (Attempt 1) yields $\rho = 0.127$ —indistinguishable from the recurrent baseline. Better architecture is insufficient when selection bias and objective mismatch dominate.
- **(N2) KL distribution supervision hurts Spearman on test.** Phase 1 Model 8 adds a KL head and improves dev Spearman to 0.640, but this does not transfer to test ($\rho = 0.127$). The KL and Huber losses jointly penalise variance, compressing output distributions (std ≈ 0.34) and suppressing rank separation.
- **(N3) NLI warm-start provides only a marginal benefit.** Switching from a generic to an NLI-initialised DeBERTa checkpoint (Attempt 3) adds only +0.008 Spearman, suggesting the NLI signal is largely overwritten during fine-tuning on 2,868 samples.
- **(N4) High per-fold OOF variance limits reliable model selection.** The final system’s OOF Spearman ranges from 0.419 (Fold 2) to 0.635 (Fold 3), a spread of 0.216. This variance means the OOF average (0.546) is an unreliable proxy for the true test performance (0.535) for any single fold configuration, and highlights the sensitivity of the task to homonym distribution.

7.5 Qualitative Analysis

We examine four representative test-set examples to illustrate the range of system behaviour. Cases are drawn from the same homonyms highlighted in the concurrent reference paper [Wu et al., 2026] to enable direct comparison.

Case 1 (Success) — *drive*, sense “hitting a golf ball off a tee.”

Precontext: *Lisa had always been competitive. Every weekend, she dedicated herself to her passion. She believed that her relentless practice would pay off someday.*

Target: *Her drive was what ultimately got her into the top university.*

Ending: *She made that long trip to show the course coordinators her dedication to going to that university, and they said that was one of the reasons why they accepted her.*

Gold: 1.0 Our prediction: 1.3 Outcome: Correct

The ending unambiguously confirms the motivational sense (“inner drive”), making the golf sense completely implausible. Our system assigns 1.3, close to gold 1.0. The NLI entailment score between story and “hitting a golf ball” is near-zero, and the ending-presence flag fires; together these scalars dominate the MLP regression.

Case 2 (Success) — *ring*, sense “a characteristic sound.”

Precontext: *John looked at his savings and smiled. He had been careful with his money for months. Now, he finally felt ready to make a big decision for their anniversary.*

Target: *He told his girlfriend he would give her a ring.*

Ending: *John was excited to finally buy the special piece of jewelry.*

Gold: 1.2 Our prediction: 1.6 Outcome: Correct

The jewelry sense is confirmed by the ending; the acoustic sense is entirely implausible. Our system correctly assigns 1.6. Both the MPNet embedding cosine similarity and the NLI entailment score between the story and “a characteristic sound” are near-zero, giving the MLP strong signal to assign a low score.

Case 3 (Error) — *shelved*, sense “delay to a later time.”

Precontext: *Emma had been collecting books her whole life. The shelves in her apartment were overflowing with volumes. She decided it was finally time to organise her collection.*

Target: *She shelved the plan to add more bookshelves.*

Ending: *She had realised she needed to put the new books somewhere and placed them neatly on the shelf.*

Gold: 1.4 Our prediction: 3.8 Outcome: Error (precontext bias)

This is the clearest failure mode: the dense library precontext strongly primes the physical-placement sense, but the target sentence uses “shelved” idiomatically to mean “postponed.” Our model assigns 3.8 because the MPNet embedding of the story text is dominated by book/library vocabulary, yielding a high cosine similarity with “delay to a later time” (both are abstract/non-physical). The ending then places books physically, contradicting the “delay” reading—but our system has no mechanism to discount the precontext when the ending signals a conflict.

Case 4 (Error) — *knitting*, sense “make textiles by knitting.”

Precontext: *Sarah sat in her armchair with her knitting needles. She was working on a warm blanket for the winter. The yarn was soft and colourful.*

Target: *Her brows were knitting together as she concentrated.*

Ending: *She furrowed her brow, deep in thought about the difficult pattern.*

Gold: 4.6 Our prediction: 1.9 Outcome: Error (ending over-anchor)

Human annotators rate the textile sense as highly plausible: the precontext establishes knitting needles and yarn so firmly that “knitting together” in the target sentence is read as a deliberate double meaning. Our model, however, over-anchors on the ending (“furrowed brow”), interpreting this as contradicting the textile sense and assigning 1.9. The NLI entailment score between the ending sentence and “make textiles by knitting” is low (furrowed brows do not entail textile production), pulling the final score down despite the strong precontext.

Systematic pattern. Cases 3 and 4 reveal a shared failure: the system lacks a learned sense of *narrative weight*—how much evidential priority

Example ated)	(abbrevi- Gold	Ours	Outcome
<i>drive</i> + “golf”; dedica- tion story; ending = univ. acceptance	1.0	1.3	Correct (Y)
<i>ring</i> + “characteristic sound”; gift-buying story; ending = jewelry	1.2	1.6	Correct (Y)
<i>shelved</i> + “delay to later time”; library context; ending = placing books	1.4	3.8	Error (pre- context bias)
<i>knitting</i> + “make tex- tiles”; blanket scene; ending = furrowed brows	4.6	1.9	Error (end- ing over- anchor)

Table 6: Qualitative summary. Full story text for each case is in §7.5.

to assign to precontext versus ending depending on story type. Structured prompting (reference system P2) partially addresses this via explicit decision rules; our architecture has no equivalent mechanism.

8 Conclusion

We present a complete two-phase system for SemEval 2026 Task 5 (AmbiStory). Phase 1 establishes that setup-grouped comparative ranking is the dominant architectural choice for recurrent models (+0.327 Spearman in a single step), but the Phase 1 best model collapses on test due to prediction-variance compression from single-split tuning and objective–metric mismatch (Model 9, $\rho = 0.611$ internal val, $\rho = 0.127$ test; §4.1). Phase 2 iterates through four attempts: a naive DeBERTa cross-encoder, grouped 5-fold CV (+0.265), NLI warm-start (+0.008), and finally an MPNet bi-encoder with explicit geometric features (+0.135) as the best system. Our final system achieves $\rho = 0.535$, Acc = 0.722, surpassing all non-LLM systems in the concurrent reference paper [Wu et al., 2026], including ELECTRA-large + LoRA ($\rho = 0.527$), while identifying narrative component weighting as the key remaining bottleneck.

Future Work. Promising directions include: (1) component-wise input structuring using separate [SEP]-delimited precontext, sentence, and ending segments to help the model allocate attention by narrative role; (2) a larger sentence encoder (e.g. MPNet-large or RoBERTa-large) to improve embedding quality; (3) uncertainty-aware isotonic

calibration for high-disagreement samples; and (4) ensemble methods combining the geometric feature approach with an LLM reranker if API access is available.

Ethics Statement

The AmbiStory dataset was collected via crowdsourced human annotation; each sample was rated by at least five annotators. We did not collect any new human-subject data and rely entirely on the publicly available AmbiStory release. The annotation process, as described in Gehring and Roth [2025], involved paid crowdworkers rating linguistic plausibility—a low-risk task with no personally identifiable information.

All experiments were run on Kaggle’s free-tier GPU allocation (Nvidia Tesla T4). No proprietary hardware or paywalled compute resources were required, lowering the barrier for replication. Code and model artefacts are publicly available at <https://github.com/SATYAJIT2747/INLP-PROJECT>.

The plausibility rating task itself is linguistically focused and does not involve generating or evaluating content that could be harmful. Potential misuse is limited; however, graded sense plausibility models could in principle be incorporated into systems that rank interpretations of ambiguous text in downstream applications (e.g. news summarisation or dialogue). Developers should ensure such downstream applications do not introduce interpretive bias against particular senses or communities.

Limitations

The final system requires GPU inference: MPNet fine-tuning per fold and the frozen NLI cross-encoder scoring take approximately 25 minutes per fold on a T4 GPU. All experiments are on English AmbiStory narratives; generalisation to other languages or longer narrative contexts is untested. Within-setup z-normalisation assumes both senses of a setup are predicted together; single-instance deployment would require a modified strategy.

A central limitation of this work is that **LLM API access was unavailable** during our experiments. The concurrent reference system [Wu et al., 2026] demonstrates that GPT-4o with structured prompting achieves $\rho = 0.731$ —a gap of 0.185 Spearman points over our best result. Closed-source inference APIs (OpenAI, Anthropic) require paid access that was not available under the

project constraints. Open-weight alternatives such as Llama-3.2-3B and Ministral-3-8B, which can be run locally or via Kaggle’s free tier, achieve substantially lower performance ($\rho = 0.134$ and $\rho = 0.472$ respectively in the reference paper), suggesting that the LLM gap is driven by model scale and structured reasoning ability rather than prompting strategy alone. Bridging this gap within a no-cost compute budget remains an open challenge.

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